Activation functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make each mathematical stage visible.

Keep score,activation,class and probability interpretation separate;show exact intermediate traces and finite-input assumptions.

Keep one habit: Assess the new chain and identity replacement with checked keys;reserve first attempts before feedback.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Trace identity **ReLU** and **sigmoid** on a numerical score

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

The score is zero and the class is 1 because equality is included.

Opening retrieval

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate a weighted score with a bias term

Reference example, not the only acceptable wording: Teacher explanation: Weighted sum 10 - 4 = 6; bias -3; score 3; **prediction** 1. Add the bias once after summing.

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

Look for: Encode a record using a fixed feature order and units

Reference example, not the only acceptable wording: Teacher explanation: A [4, 2], B [2, 4], C [3, 3]. Each has dimension 2. The table has 3 rows and 2 columns.

Pictured end goal

By the end, I can explain **activation function**s using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Bias terms and decision boundaries

Representing inputs as vectors

For the compact alternative schedule, use two 45-minute sessions to compare **activation function**s before network **training**. Students should trace z, activation a and any later **output** separately. **Sigmoid** arithmetic may use the supplied approximations.

Focus on the change in function shape and the distinction between an activation, class decision and probability claim.

Outcomes

Trace identity **ReLU** and **sigmoid** on a numerical score

Explain how a nonlinear activation changes a computation

Distinguish an activation **output** from a final class or verified probability

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach transformations and decisions as distinct steps

> Use the same incoming scores before discussing the composed chain.

Main definitions,numerical interpretation and guided key

First compute an affine score z = w dot x + b. An activation then returns a = f(z). In a network this value may become an **input** to another layer; it is not automatically a class **prediction**.

We will compare three functions using the same incoming scores so that only the activation changes.

Identity returns z unchanged. **ReLU**, pronounced ree-loo, returns the larger of zero and z. Thus **ReLU**(-2) = 0, **ReLU**(0) = 0 and **ReLU**(2) = 2. Positive scores keep their magnitude; **ReLU** does not turn every positive value into 1.

These are fixed arithmetic functions. Selecting a function or changing its **input** is not itself a **training** procedure. Later weeks will introduce losses and parameter updates.

Sigmoid is 1 / (1 + exp(-z)). Here exp(t) means e raised to power t, where e is approximately 2.718. A calculator or the supplied Python code can evaluate it; no calculus is required this week.

For z = -2, 0 and 2, **sigmoid output**s are approximately 0.1192, 0.5 and 0.8808.

For every finite mathematical **input**, **sigmoid** is strictly between 0 and 1, though computer rounding can produce an endpoint for extreme scores.

Negative **input**s produce values below 0.5, zero gives 0.5, and positive **input**s give values above 0.5. A class rule still needs to be stated: if class 1 means **sigmoid**(z) >= 0.5, these three **prediction**s are 0, 1 and 1.

The two algebraically equivalent branches avoid taking exp of a large positive number. A value within [0, 1] is not automatically a reliable probability: its interpretation depends on the fitted task and evaluation **evidence**.

Incoming score z · Identity z · ReLU max(0, z) · Sigmoid ≈ · Class if sigmoid≥0.5

> **Incoming score z:** -2
>
> **Identity z:** -2
>
> **ReLU max(0, z):** 0
>
> **Sigmoid ≈:** 0.1192
>
> **Class if sigmoid≥0.5:** 0

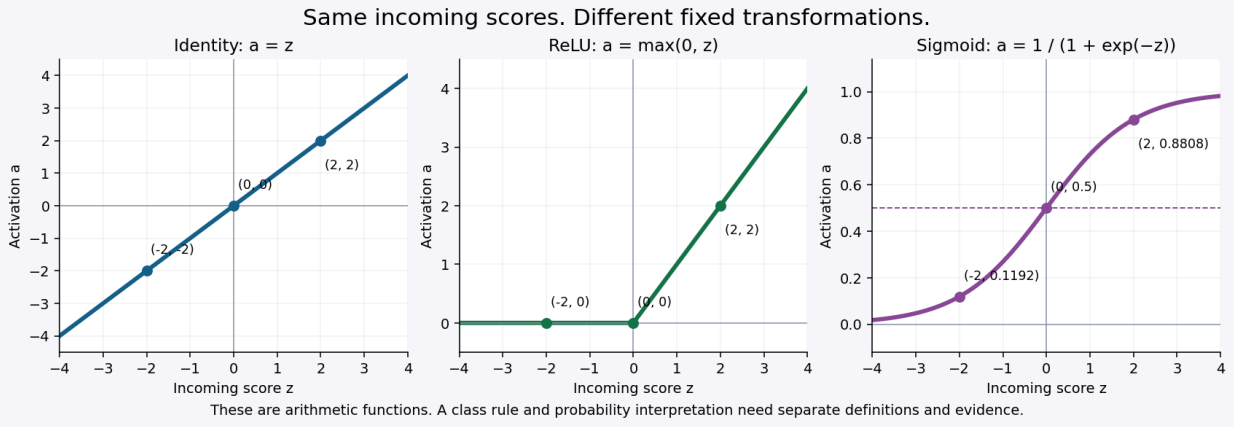
> **Incoming score z:** 0
>
> **Identity z:** 0
>
> **ReLU max(0, z):** 0
>
> **Sigmoid ≈:** 0.5
>
> **Class if sigmoid≥0.5:** 1

> **Incoming score z:** 2
>
> **Identity z:** 2
>
> **ReLU max(0, z):** 2
>
> **Sigmoid ≈:** 0.8808
>
> **Class if sigmoid≥0.5:** 1

Guided: identity **output**s -1 and 1; **ReLU output**s 0 and 1. **Sigmoid** class **prediction**s under >= 0.5 are 0 and 1. Exit: the computation establishes an activation of about 0.8808 for score 2.

It does not establish calibrated probabilities, correctness on this individual case or reliability on future populations.

Labelled function curves



Complete pictured chain and scope

See what **ReLU** removes before the next weighted step.

Toy network z=x-3; h=max(0, z); y=3h+2. **Input**s x=1, 3, 4.

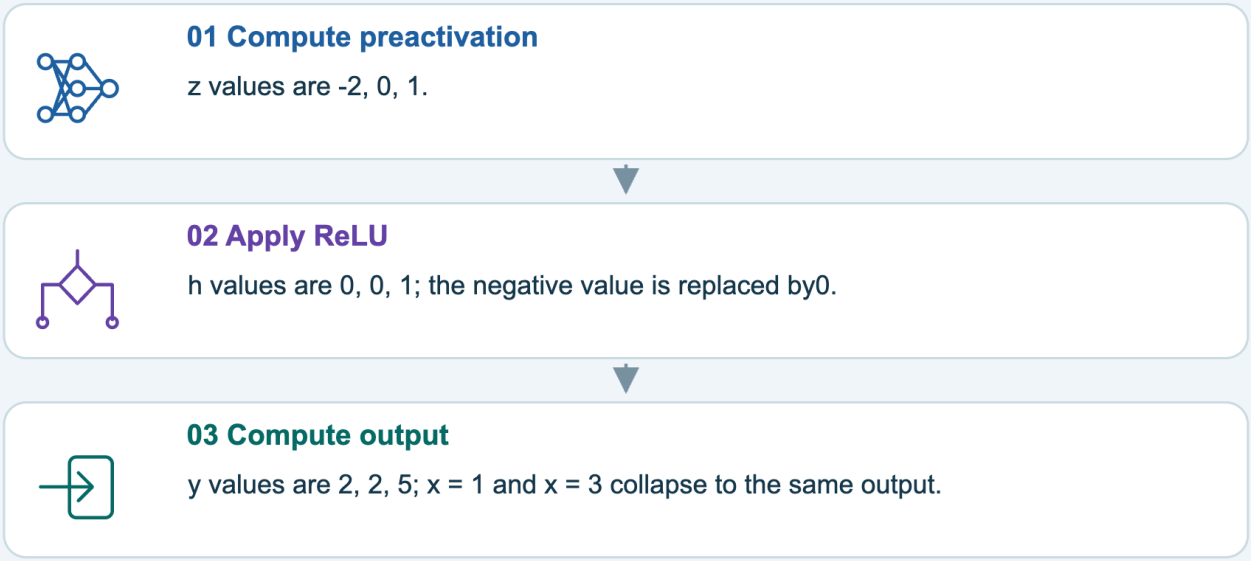
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses z=x−3, h=ReLU(z), y=3h+2. Inputs1/3/4 give z−2/0/1, h0/0/1, y2/2/5. Inputs1 and3 have become indistinguishable to this final step because both pass h0. Replace only ReLU with identity:the same z values survive as h−2/0/1, y−4/2/5. This is not the independent chain, which supplies different constants and inputs. No fitting or observed real-world success is represented.

Trace ReLU and the identity replacement

AI ACADEMY / AI-5/4 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute preactivation	z values are -2, 0, 1.
2. Apply ReLU	h values are 0, 0, 1; the negative value is replaced by0.
3. Compute output	y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

ReLU means keep a positive value and replace a negative one with zero. The **output** layer receives that transformed value, not the original **input**.

Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

ReLU is one activation choice. Clipping negatives does not itself train the network or prove useful representation.

Change: Replace **ReLU** with identity.

Prediction: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Reason: Identity retains the negative intermediate value instead of clipping it.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Activation function	A function applied to a computed score
ReLU	The function max(0, z), retaining positive scores and replacing negative scores with zero
Sigmoid	The function 1 / (1 + exp(-z)), mapping finite real scores into the interval between 0 and 1
Nonlinear function	A function that cannot be represented by one fixed slope and offset over its entire domain

Three functions receive the same score

SAME INCOMING SCORE	APPLY ONE FUNCTION	ACTIVATION OUTPUT
z is -2	Identity of z	-2
z is -2	ReLU of z	0
z is -2	Sigmoid of z	About 0.1192

Each row starts again from the same score -2 and applies one function. The functions are alternatives here, not successive steps. These outputs are activation values, not class labels.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Three functions receive the same score

SAME INCOMING SCORE	APPLY ONE FUNCTION	ACTIVATION OUTPUT
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z is -2	ReLU of z	0
z is -2	Sigmoid of z	About 0.1192

Each row starts again from the same score -2 and applies one function. The functions are alternatives here, not successive steps. These outputs are activation values, not class labels.

For z = -1 and z = 1, give identity and ReLU outputs. Using sigmoid(-1) = 0.2689 and sigmoid(1) = 0.7311, give classes under the rule a >= 0.5.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Activation functions

The facts we will use

AI ACADEMY / AI-5/4

ReLU hidden computation at x=1

Input 1 x weight 1 = 1

Add the products and bias -3: total = -2

ReLU: use max(0, total) -> output 0

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/4

Activation functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute preactivation

z values are -2, 0, 1.



2. Apply ReLU

h values are 0, 0, 1; the negative value is replaced by0.



3. Compute output

y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

CHANGE ONE THING

Replace ReLU with identity.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

Teacher check: Teacher explanation: Classes are 0, 1, 1. **Sigmoid**(0) equals 0.5 and the declared >= comparison includes equality.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Activation function**s. Proposed explanation: “**ReLU** returns a class label”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Positive scores retain magnitude, so **ReLU**(2) is 2. A final class rule is separate.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

x=1 and x=3 both give h = 0 and y = 2. After **ReLU**, the downstream step receives identical values for those **input**s. Changed-case check: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Identity retains the negative intermediate value instead of clipping it.

Changed-case reasoning: h becomes -2, 0, 1 and **output**s become -4, 2, 5. Identity retains the negative intermediate value instead of clipping it.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use a number line and a three-column z, a, y table. Supply **sigmoid** values rather than requiring exponent calculations from memory.

Extension: use several **input**s to verify that **sigmoid**(z) and **sigmoid**(-z) sum to 1 within rounding error, while noting that this symmetry is not **evidence** of calibration.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite boundary example. Introduce z as the score and a as the transformed value.
7 to 18 minutes	Trace identity and ReLU for -2, 0 and 2. Complete W1, checking that positive values keep magnitude.
18 to 32 minutes	Introduce the sigmoid formula and supplied values. Trace or run the stable code; complete W2 using the explicit equality policy.
32 to 45 minutes	Compare the visual rows and complete the guided -1 and 1 case. Keep activation values separate from classes.
Session 2 start to 8 minutes	Retrieve ReLU at one negative and one positive input. Ask whether one line covers both sides.
8 to 20 minutes	Complete W3 and trace the two worked chains at x = -1 and x = 1. Show the identity-chain simplification by substitution.
20 to 35 minutes	Collect W4 independently, requiring a three-stage table. Do not award full credit for final y values without the intermediate trace.
35 to 45 minutes	Complete W5 and the exit response. Use the fresh retry after feedback; connect to composing layers next week.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Same-score table and probes as supplied. Finite extremes avoid overflow but can round to0/1. Do not infer calibrated reliability from sigmoid range. Independent x0/2/5:z–2/0/3, a0/0/3, y1/1/7; identity gives–3/1/7. Inputs0 and2 collapse through ReLU. W3 midpoint0 differs from endpoint-average 1; W5 asks actual calibration/evaluation evidence, not a range argument.

Diagnose the current misconceptions

ReLU returns a class label: Positive scores retain magnitude, so **ReLU**(2) is 2. A final class rule is separate.

A bounded number is a calibrated probability: A range restriction comes from the formula. Calibration needs appropriate **evidence**.

Nonlinearity guarantees a better result: It changes which functions can be represented; data, **fitting** and valid evaluation still determine model quality.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Let z = x - 2, a = **ReLU**(z), y = 2a + 1. For x = 0, 2, 5, record every z, a and y. Then replace **ReLU** with identity and give all three y values. Which **input**s have the same **ReLU output**, and why?

Teacher explanation: For x = 0, 2, 5: z = -2, 0, 3; a = 0, 0, 3; y = 1, 1, 7. Identity gives y = -3, 1, 7. **Input**s 0 and 2 both give a = 0 because **ReLU** maps negative and zero scores to zero.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace three activations

For scores -2, 0, 2, list identity and **ReLU output**s. Explain why **ReLU**(2) is not 1.

Teacher explanation: Identity gives -2, 0, 2; **ReLU** gives 0, 0, 2. **ReLU** retains the positive score itself; it is not a step classifier.

W2 Apply a declared class rule

Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

Teacher explanation: Classes are 0, 1, 1. **Sigmoid**(0) equals 0.5 and the declared >= comparison includes equality.

W3 Check the midpoint

Compare **ReLU** at **input**s -2, 0, 2. Give the average of the two endpoint **output**s and the **output** at the midpoint; explain what this shows.

Teacher explanation: Endpoint **output**s 0 and 2 average to 1; the midpoint **input** 0 produces 0. The mismatch shows **ReLU** is not one affine function over these **input**s.

W5 Limit a probability claim

A model returns a **sigmoid** value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

Teacher explanation: No. It establishes only the computed activation value.

Probability interpretation requires a suitable fitted model and evaluation, including checking predicted probabilities against outcome frequencies on relevant separate data; one value does not establish calibration.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: identity **output**s -1 and 1; **ReLU output**s 0 and 1. **Sigmoid** class **prediction**s under >= 0.5 are 0 and 1. Exit: the computation establishes an activation of about 0.8808 for score 2.

It does not establish calibrated probabilities, correctness on this individual case or reliability on future populations.

W1 Trace three activations

W2 Apply a declared class rule

W3 Check the midpoint

W4 Trace a new chain independently

Let z = x - 2, a = **ReLU**(z), y = 2a + 1. For x = 0, 2, 5, record every z, a and y. Then replace **ReLU** with identity and give all three y values. Which **input**s have the same **ReLU output**, and why?

Teacher explanation: For x = 0, 2, 5: z = -2, 0, 3; a = 0, 0, 3; y = 1, 1, 7. Identity gives y = -3, 1, 7. **Input**s 0 and 2 both give a = 0 because **ReLU** maps negative and zero scores to zero.

W5 Limit a probability claim

Independent assessment and fresh retry

Assess W4 for z values -2, 0, 3, **ReLU** values 0, 0, 3, final **output**s 1, 1, 7 and identity **output**s -3, 1, 7. Fresh retry: z = 2x - 3, a = **ReLU**(z), y = a - 1 at x = 1, 1.5, 3.

The triples (z, a, y) are (-1, 0, -1), (0, 0, -1), (3, 3, 2). Require an explanation of which distinctions were erased.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **activation function**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace identity **relu** and **sigmoid** on a numerical score.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use a number line and a three-column z, a, y table. Supply **sigmoid** values rather than requiring exponent calculations from memory.

Extension: use several **input**s to verify that **sigmoid**(z) and **sigmoid**(-z) sum to 1 within rounding error, while noting that this symmetry is not **evidence** of calibration.

Week 5 composes layers and traces how activations become **input**s to the next computation.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/5, ai-5/7, ai-5/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Trace identity **ReLU** and **sigmoid** on a numerical score

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

Reasoned practice answer: Teacher explanation: Classes are 0, 1, 1. **Sigmoid**(0) equals 0.5 and the declared >= comparison includes equality.

Check every learner before discussion: Use the worked case for **Activation function**s. Proposed explanation: "**ReLU** returns a class label". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Positive scores retain magnitude, so **ReLU**(2) is 2. A final class rule is separate.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace **ReLU** with identity. Require a **prediction** and a reason before comparing. Expected result: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Reason: Identity retains the negative intermediate value instead of clipping it.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Activation functions: the case

AI ACADEMY / AI-5/4 · 01 · LOOK AT THE FACTS

Activation functions: the case

The situation

See what ReLU removes before the next weighted step.

Supplied evidence

Toy network z=x-3; h=max(0, z); y=3h+2.
Inputs x=1, 3, 4.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain activation functions using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

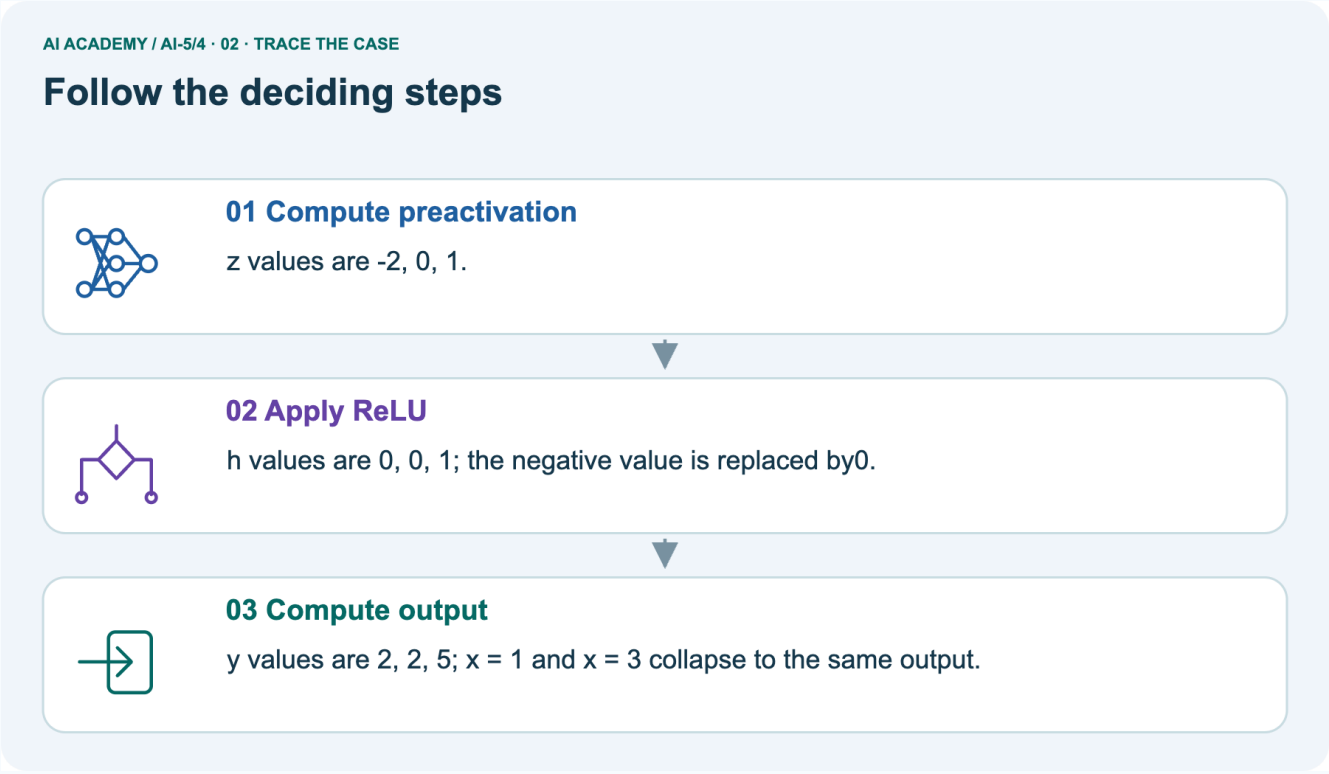
See what **ReLU** removes before the next weighted step. Supplied facts: Toy network z=x-3; h=max(0, z); y=3h+2. **Input**s x=1, 3, 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



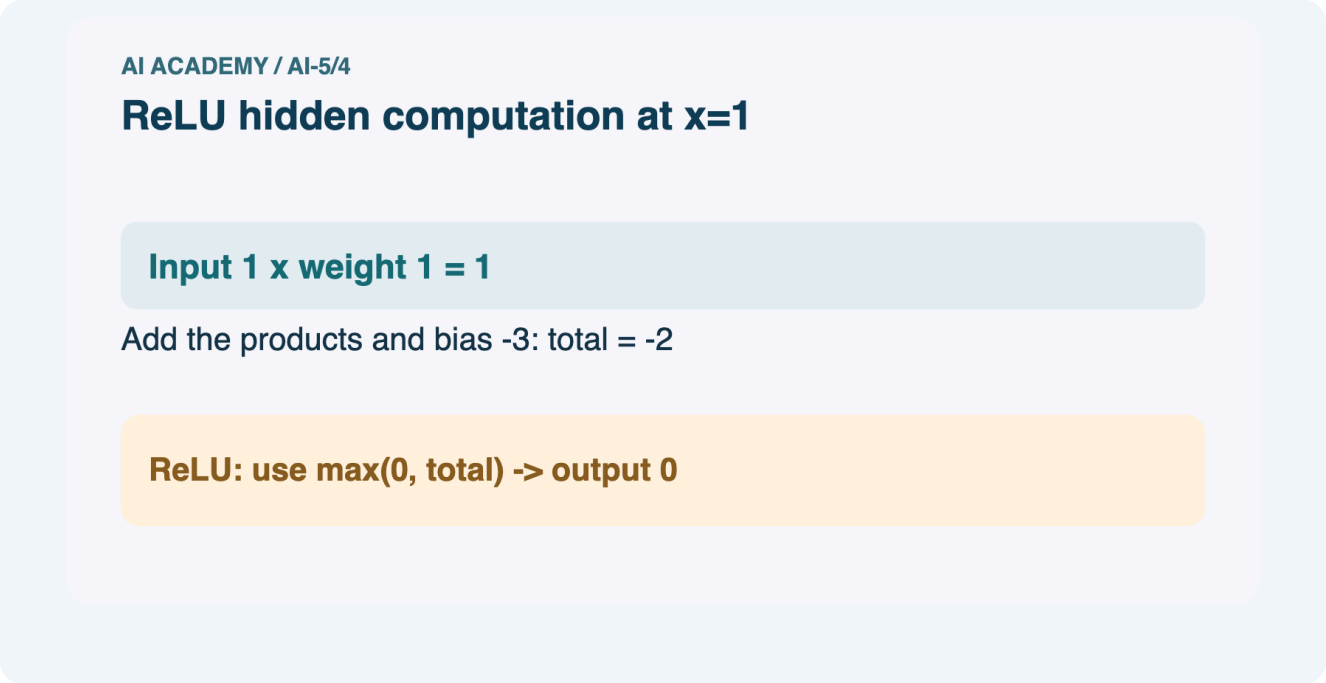
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute preactivation: z values are -2, 0, 1. → Apply **ReLU**: h values are 0, 0, 1; the negative value is replaced by0. → Compute **output**: y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. ReLU hidden computation at x=1



Follow the labelled picture with the matching explanation. Open the vector diagram

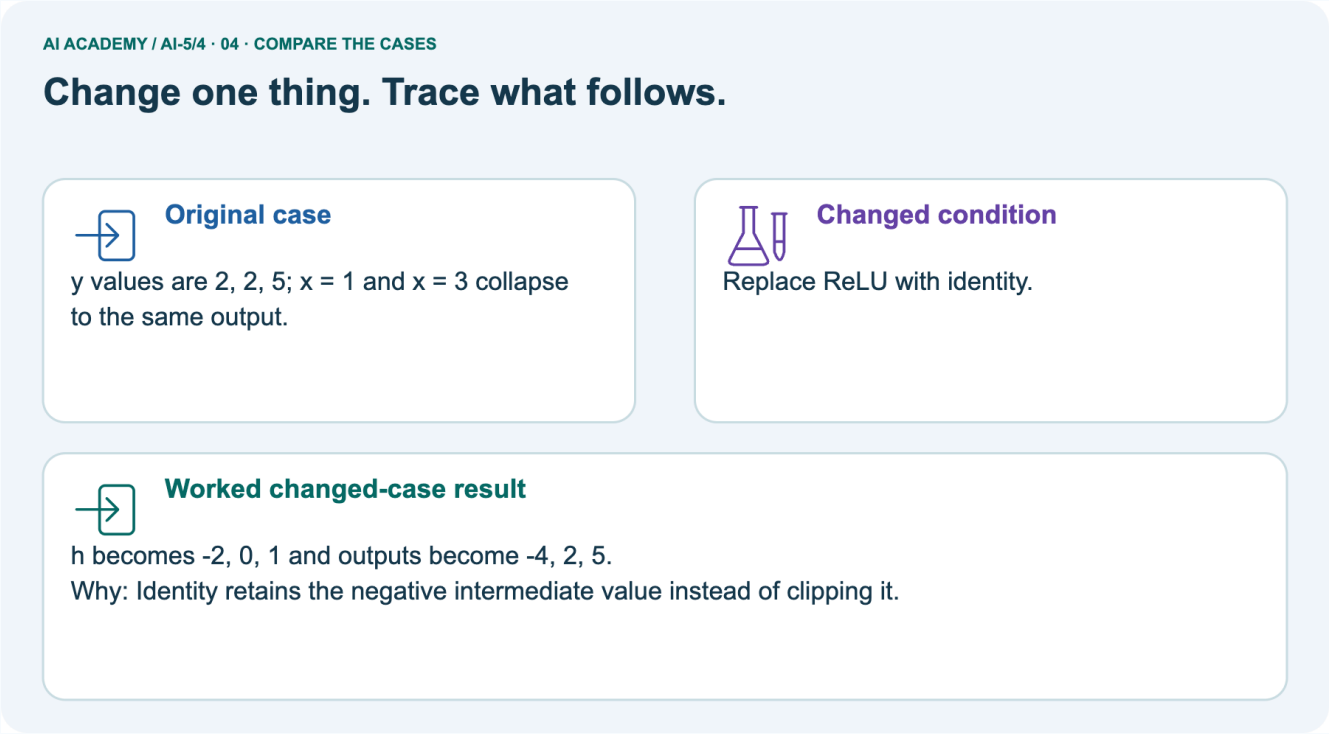
ReLU hidden computation at x=1. **Input**s: [1]; weights: [1]; bias: -3; activation: **relu**; **output**: 0.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**. Changed condition: Replace **ReLU** with identity. Expected result: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

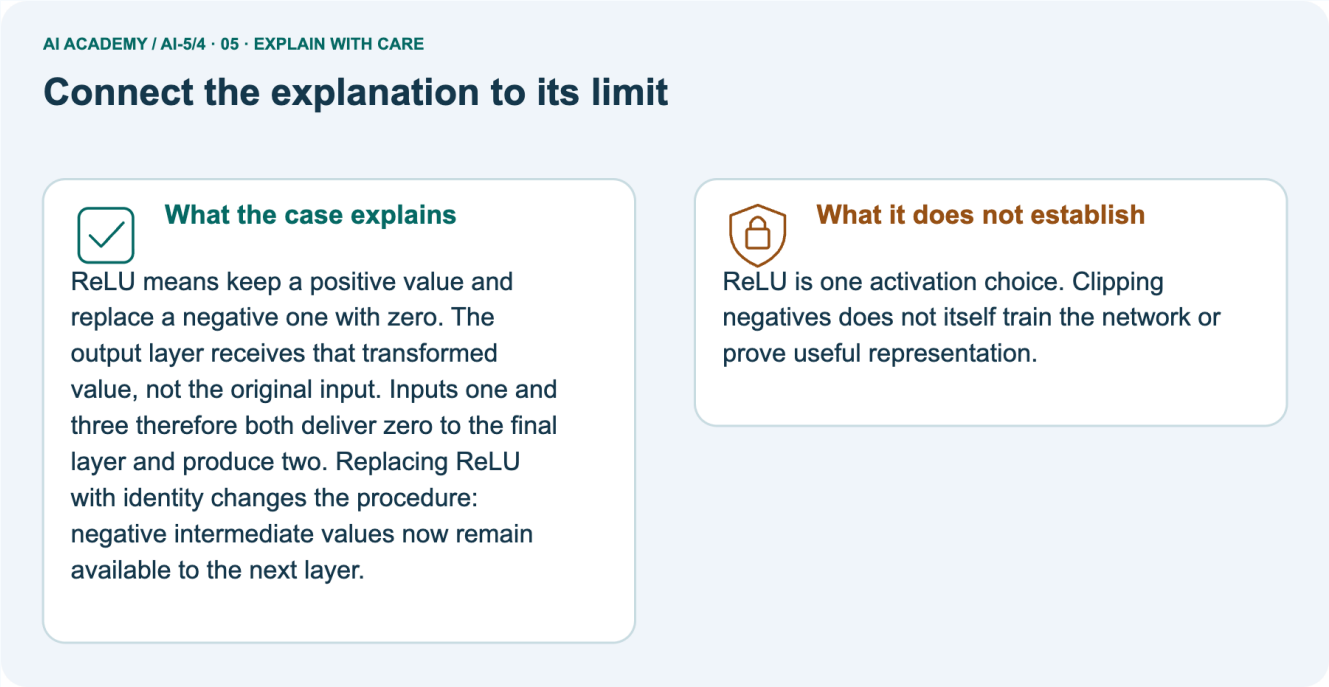
Reason: Identity retains the negative intermediate value instead of clipping it.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Identity retains the negative intermediate value instead of clipping it.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

ReLU means keep a positive value and replace a negative one with zero. The **output** layer receives that transformed value, not the original **input**.

Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

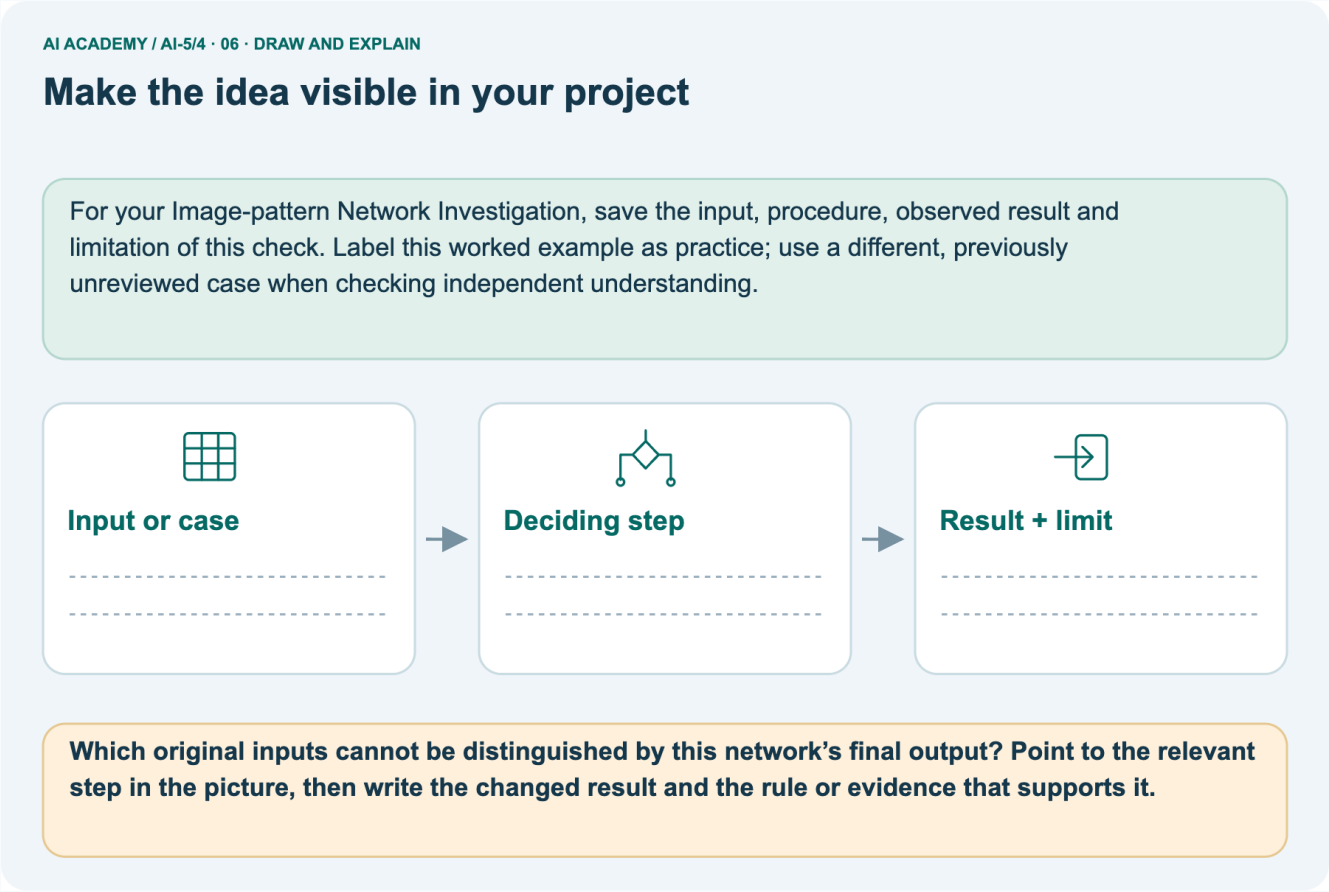
Limit: **ReLU** is one activation choice. Clipping negatives does not itself train the network or prove useful representation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **ReLU** is one activation choice. Clipping negatives does not itself train the network or prove useful representation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: x=1 and x=3 both give h = 0 and y = 2. After **ReLU**, the downstream step receives identical values for those **input**s. Changed-case check: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Identity retains the negative intermediate value instead of clipping it.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Activation functions: the case

02. Follow the deciding steps

03. ReLU hidden computation at x=1

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.